BT3041 – Analysis and Interpretation of Biological Data

Assignment 1

Instructions

Please provide your analyses results in a report form, specifically answering each of the questions with relevant plots, etc. Also, do state any assumptions made clearly in the report. Attach the Google Drive link to your software code (Python) used to perform the calculations for the report.

Submit your completed assignment report and supplementary files (if any) in Turnitin (Link for submission will be shared separately) by 08/03/2026 13:00:00. Any late submissions beyond this timeline will incur a penalty of 10 marks per hour. Obviously, if you submit it later than 10 hours, the entire assignment is considered forfeited.

You are provided with a dataset from a study that tracked mice (the NOD/ShiLtJ strain) to see how they develop Type 1 Diabetes as they grow. The study monitored the mice between the ages of 3 and 30 weeks, checking their body weight and blood glucose levels.

You can download the data from the Moodle page.

Perform the analysis mentioned below on the given data (Total Marks = 50):

1. Create separate histograms to visualize the blood glucose distribution for female and male mice across different weeks, with each graph showing the distributions of male and female mice as individual overlaid or side-by-side distributions. Based on the symmetry, interpret whether these features follow a normal distribution or not (5 marks).

2. Construct boxplots illustrating blood glucose distributions for male mice across early age (weeks 3, 5, 7) and later age (weeks 14, 15, 16) phases positioned side-by-side. Also construct a similar boxplot for female mice. Provide commentary on differences in medians and interquartile ranges observed between age phases within each sex (5 marks).

3. Using the blood glucose dataset, compute Z-scores using two approaches: (i) across mice within each individual week, and (ii) across weeks within each individual mouse. Generate separate heatmaps to visualize and compare these Z-score matrices, then provide observations on patterns and trends revealed by each standardization method. (5 marks).

4. Refer to the Clinical Chemistry sheet and create a scatter plot and assess if any of the following two measurements have any correlation: (a) Blood glucose level (GLU) against High Density Lipoprotein (HDLX) of mice on week 30, (b) Blood glucose level (GLU) against Alanine Transaminase (ALT) of mice on week 30, (c) Serum Albumin level against High Density Lipoprotein (HDL) of mice on week 30, (d) Blood glucose level (GLU) against Creatine Kinase (CK) of mice on week 30 and (e) Total Protein (TP) against Blood Urea Nitrogen (BUN) of mice on week 30. Additionally, compute the covariance in all the above conditions and provide inferences based on these values (15 marks).

6. Using an appropriate hypothesis testing method, evaluate whether the mean of fat tissue mass significantly increased from week 8 to week 30 in male mice fed a 6% fat diet. (10 marks).

7. Evaluate whether there is a significant difference in WBC counts between male and female mice on week 30. Choose the test by plotting a histogram or density plot (10 marks).